1, Rings and ideals

Ring: $(R, +, \cdot)$
- $(R, +)$ abelian group
- $(R, \cdot)$ semigroup
- distributive law

All our rings will have a unit $1 \in R$
All our rings unless specified will be commutative

$0 \in R$ neutral element under $+$
The trivial ring $\{0\}$ has $1 = 0$
$u \in R$ is a unit (= invertible element) if there exist $\exists v \in R.\ uv = vu = 1$. This v is unique $v = u^{-1}$

$R^\times$ or R^{-1} or $GL(R)$... units of R

Ring homomorphism is a map $\varphi: R \to R'$ that preserves sums, product and ones

...

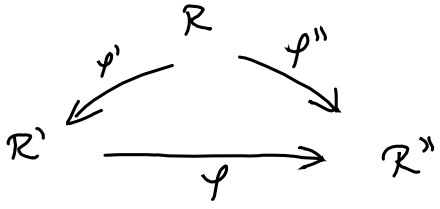
R-algebras: ring R' with a homomorphism

$\varphi : R \to R'$

called structure homomorphism

Every ring is a $\mathbb{Z}$-algebra

R-algebra homomorphism between R', R'' is a ring homomorphism that is compatible with structure maps



Examples:

- $\mathbb{Z}_2$
- R ring, Y set

  $R^Y = \{ Y \to R \}$ is a ring with point wise operations $f, g \in R^Y$ $(f+g)(y) = f(y) + g(y)$

  $(f \cdot g)(y) = f(y) \cdot g(y)$

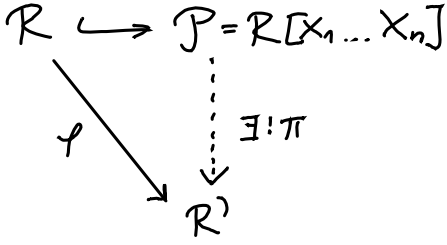
  identity: $y \mapsto 1$

Ring of polinomials: $\mathcal{P} = R[X_1, \dots X_n]$

$\mathcal{P}$ has the following universal property (UL):

$\forall$ hamamorphism $\varphi : R \to R'$. $\forall a_1, \dots a_n \in R$

$\exists!$ hamamorphism $\pi : \mathcal{P} \to R'$ s: $\pi_{|R} = \varphi \wedge \pi(X_i) = a_i$



$$\pi\left(\sum_i r_i X_1 \cdots X_n\right) = \sum_i \varphi(r_i) a_1 \cdots a_n$$

• Ideal: $I \trianglelefteq R$ $\quad 0 \in I$

$x, y \in I \Rightarrow x+y \in I$

$r \in R, x \in I \Rightarrow rx \in I$

$S \subseteq R$ $\quad \langle S \rangle$ ... Ideal of R generated by S

$\langle S \rangle = \{ \sum_{s \in S}^{\text{finite}} a_s s \; ; \; a_s \in R \}$

$\langle a \rangle$ ... principal ideals

Operations on ideals

• I_λ $(\lambda \in \Lambda)$; ideals of R

$$\sum_{\lambda \in \Lambda} I_\lambda = \{ \sum_{\lambda \in \Lambda}^{\text{finite}} b_\lambda \; ; \; b_\lambda \in I_\lambda \}$$

• $\cap I_\lambda \trianglelefteq R$ $\qquad \bigcap_{\emptyset} I_\lambda = R$

• $\prod I_\lambda = \langle \{ \prod_{\lambda \in \Lambda'} b_\lambda \; ; \; \Lambda' \overset{\text{finite}}{\subseteq} \Lambda, \; b_\lambda \in I_\lambda \} \rangle$

• $(I : J) = \{ x \in R \; ; \; xJ \subseteq I \}$ transporter

• $I \cdot J \subseteq I \cap J \subseteq I, J \subseteq I + J$

$I \subseteq (I : J)$

$I(J+K) = I \cdot J + IK$ DN

$I \trianglelefteq R$ is proper if $I \subsetneq R$

$I \trianglelefteq R$ is trivial if $I = \{0\}$

$\langle x \rangle = R \Leftrightarrow x \in R^\times$

($\Leftarrow$) obvs

($\Rightarrow$) $\langle x \rangle = R \Rightarrow \exists u \in R \quad ux = 1 \Rightarrow x \in R^\times$

$\varphi : R \to R' \qquad I \lhd R$

$\varphi(I)$ tipically not ideal (

$\mathbb{Z} \hookrightarrow \mathbb{Q}$

$\uparrow \quad \nearrow$

$2\mathbb{Z}$

$\langle \varphi(I) \rangle =: I^e$ "estention of I"

$I' \lhd R'$

$\varphi^*(I') \lhd R$ "kontraction of I'" $I'^c = \varphi^*(I')$

$\varphi : R \to R' \Rightarrow \ker \varphi = \varphi^*(0) \lhd R$

$\ker \varphi = \{0\} \Leftrightarrow \varphi$ is injective

$I \trianglelefteq R$ Form cosets $\{x+I ; x \in R\} = R/I$

$$(x+I)+(y+I) = (x+y)+I)$$
$$(x+I)(y+I) = xy+I$$

Quotient or factor ring

$$K : R \longrightarrow R/I \qquad x \mapsto x+I$$

canonical quotient homomorphism

K is surjective

$\ker K = I$

$$\boxed{R/\ker\varphi \cong \operatorname{im}\varphi}$$

$R \quad I \trianglelefteq R \quad I \subseteq J \trianglelefteq R \quad K : R \to R/I$

$$J/I = \{y+I \mid y \in J\} = K(J) \trianglelefteq R/I$$

maps $J \mapsto J/I$ in $J' \mapsto K^{-1}(J)$ are inverse hence bijection

$$\frac{R/I}{J/I} \cong R/J$$

Proposition: $\mathcal{P} = R[X]$; $a \in R$. Let $\pi: \mathcal{P} \to R$ be the R-ring homomorphism $X \mapsto a$

a) $\ker \pi = \{F \in \mathcal{P};\ F(a) = 0\} = \langle x - a \rangle$

b) $\mathcal{P}/\ker \pi \cong R$

Proof: $G := X - a$

$\forall F \in \mathcal{P}.\ \exists H \in \mathcal{P}.\ \exists r \in R\ \ni: F = GH + r$

B; linearity it suficies to prove this for $f = X^n$

$f = X^{n-1}X = X^{n-1}(G + a) = GX^{n-1} + aX^{n-1}$

Repeat this starting with X^{n-1}

a) $\pi(F) = \underset{0}{\pi(G)}\,\pi(H) + \underset{r}{\pi(r)} = r$

Hence $F \in \ker(F) \Leftrightarrow r = 0 \Leftrightarrow F(X-a) \cdot H$

$\Leftrightarrow F \in \langle X - a \rangle$

b) $\mathcal{P}/\ker \pi \cong \operatorname{im} \pi = R$

■

$\deg F = \max \{\deg(M);\ M \text{ monomial in } F\}$

$M = X_1^{\alpha_1} \dots X_n^{\alpha_n}$

$\deg(M) = \sum \alpha_i$

$\deg(0) = -\infty$

$\deg(F \cdot G) \leq \deg F + \deg G$

Idempotents if $e^2 = e$

Poglej si pri Filipu, če je potrebno

Prop: Let R be a ring and e', e'' complementary idempotents ($e''=1-e'$). Set $R'=Re'$ in $R''=Re''$

Define $\varphi: R \longrightarrow R' \times R''$

$$x \longmapsto (xe', xe'')$$

Then R is isomorphiseen $R' \cong R/Re''$, $R'' \cong R/Re'$

Proof:

Consider $\varphi': R \longrightarrow R$

$$x \longmapsto xe'$$

φ' is homomorphism $\varphi'(xy) = xye' = xye'^2 = \varphi(x)\varphi(y)$

$\ker \varphi' = R''$

$(\subseteq)$ $\varphi'(x)=0 \Rightarrow xe'=0$

$$x \cdot 1 = x(e'+e'') = 0 + xe'' \in R''$$

$(\supseteq)$ $\varphi(xe'') = x\underbrace{e''e'}_{=0} = 0$

Then $\operatorname{im}\varphi = R'$ and $R' \cong R/Re''$

Similarly we have $\varphi'': x \longmapsto xe''$

Then $\varphi = (\varphi', \varphi'')$ is homomorphism

$\ker \varphi = 0$

$x \in \ker\varphi \Rightarrow xe'=0$ and $xe''=0$

$\Rightarrow 0 = x(e'+e'') = x \cdot 1 = x$ ✓

$\operatorname{im}\varphi = R' \times R''$

$(xe', x''e'') = \varphi(x'e' + x''e'')$

$\Rightarrow \varphi$ is isomorphism

■

II Prime ideals

$x \in R - \{0\}$ is a zero devisor $\Leftrightarrow \exists y \in R . xy = 0$

$zdiv R = \{ x \in R \text{ zero devisor} \}$

Def $S \subseteq R$ is multiplicative if $1 \in S$ & $S \cdot S \subseteq S$

$P \trianglelefteq R$ is prime if $R - P$ is multiplicative

Eg: $P \trianglelefteq R$ is prime

Ring is a field if $0 \neq 1$ and an every non zero element is invertable

Ring where $\langle 0 \rangle$ is prime ideal is integral domain (od. helobar)

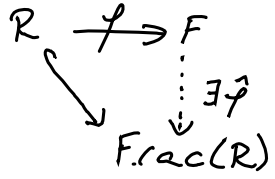
such ring has no zero devisors

- R integral domain $\Rightarrow$ can form a field of fractions

$$Frac(R) = \left\{ \frac{a}{b} ; a \in R ; b \in R - \{0\} \right\}$$

- every integral domain is a subring of a field
- every subring of a field is integral

UL for Frac(R): Each embeding of R into F Field extends to Frac(R) $\xrightarrow{\hat{\varphi}}$ F



$\hat{\varphi}\left(\frac{a}{b}\right) = \frac{\varphi(a)}{\varphi(b)}$

- Polynomial rings over integral domains

  R domain, $P = R[x]$

  P is also a domain and $\deg(FG) = \deg F + \deg G$

  We order P with so-called graded lexicographic (grlex) ordering:

  order first by degree, then lexicografically

  $x_1^4 x_2 < x_1^3 x_2^3 x_3 < x_1^2 x_2 x_3^4$

  Among all the monomials pick the largest one. This is LM(F) = leading monomial of F

  LT(F) = leading term of F $a \cdot LM(F)$ if $a \cdot LM(F)$ appears in F

  LC(F) = leading term of F a

$F, G \in P \Rightarrow LT(FG) = LT(F) \cdot LT(G)$

$LT(F) = a \cdot m$, $LT(G) = a \cdot n$

Then $a \cdot b \neq 0$ bcos R is a domain

$\deg(m \cdot n) = \deg(m) + \deg(n) = \deg(F) + \deg(G)$

if $m' \neq m$ or $n' \neq n$ then $m'n' <_{grlex} mx$ by max of m, n !

Hence $LT(F \cdot G) = abmn = LT(F) \cdot LT(G)$

P domain $\Rightarrow$ hen form $Frac(P) =: K(X)$,

$K = Frac(R)$ ↑ field of rational functions

$P^\times = R^\times$

if $F \cdot G = 1$ then $\deg(FG) = 0$

$\deg(F) + \deg(G) \geq 0$

lema: let $\varphi: R \to R'$ is homomorphism and $T \subseteq R'$.

a) if T is multiplicative then $\varphi(T)$ is too

b) if φ is surjective (onto), then also the converse holds

Proof:

a) $S = \varphi^{*}(T)$

Since $\varphi(1)=1 \in T$ we have $1 \in S$

For $x,y \in S$ we have

$$\varphi(xy) \in \varphi(S) \subseteq T$$
$$\| $$
$$\varphi(x)\cdot\varphi(y) \in \varphi(S)\cdot\varphi(S) = T\cdot T$$
$$\Rightarrow T\cdot T \subseteq T$$

b) Now assume $S = \varphi^{*}(T)$ is multiplicative

T is multiplicative

- $1 \in T$ since $1 \in S \wedge \varphi(1)=1$
- given $x', y' \in T$ then $\exists x, y \in S$ $x' = \varphi(x)$, $y' = \varphi(y)$
  Then $x\cdot y \in S$ and $\varphi(x,y) = \varphi(x)\,\varphi(y) = x'y' \in T$

Corollary:

$\varphi: R \to R'$ hom, $J \trianglelefteq R'$. $I := \varphi^*(J)$

a) if J is a prime ideal then so is I

b) if φ is onto (surj) $\wedge$ I is prime $\Rightarrow$ J is prime

Proof: Apply the lemma to the complement of J

Corollary: Let $P \trianglelefteq R$. Then

P prime $\Leftrightarrow$ R/P is a domain

Definition: $M \trianglelefteq R$ is a maximal ideal of $M \subsetneq R$ and $\nexists I \trianglelefteq R$. $M \subsetneq I \subsetneq R$

Proposition:

Ring R is a field $\Leftrightarrow$ (0) is a max ideal

Proof:

($\Rightarrow$) $\exists\, 0 \neq x \in I$. Then $\exists x^{-1} \in R$ Hence $1 = x \cdot x^{-1} \in I$

So $I = R$

($\Leftarrow$)

Pick $x \in R - (0)$ then $(0) \subsetneq \langle x \rangle \lhd R$

Since (0) is a max ideal $\langle x \rangle = R$

in particular $1 \in \langle x \rangle$ so $\exists y \in R.\ xy = 1$

So x is invertable

Coro. Let R be a ring. $M \lhd R$. Then M is max ideal $\Leftrightarrow$ R/M is a field

Proof: M is a max ideal $R \Leftrightarrow (0)$ is a max ideal R/M $\overset{\text{prop}}{\Leftrightarrow}$ R/M is a field ∎

Cor. Every max ideal is prime

Proof Every field is an int domain

Examples:

a) R int domain; $R[x,y] =: \mathcal{P}$. Then $\langle x \rangle \triangleleft \mathcal{P}$ is a prime ideal that is not maximal

$$\langle x \rangle \subsetneq \langle x, y \rangle \subsetneq \mathcal{P}$$

$\langle x \rangle$ is prime since $\mathcal{P}/\langle x \rangle = R[y]$ is a domain

$\langle x, y \rangle$ is max $\Leftrightarrow$ $\mathcal{P}/\langle x, y \rangle \cong$ is a field

b) R ring, $\mathcal{P} = R[x_\lambda ; \lambda \in \Lambda]$. For each $\lambda \in \Lambda$

$$M := \langle x_\lambda - a_\lambda ; \lambda \in \Lambda \rangle$$

Then $\mathcal{P}/M \cong R$

Theorem: Every proper ideal is contained in a maximum ideal

Proof:

$I \subsetneq R$

$\mathcal{S} = \{ J \subsetneq R ; I \subseteq J \}$

We want to use Zorn lema

- $I \in \mathcal{S} \Rightarrow \mathcal{S} \neq \emptyset$
- every chain has an upper bound

  chain $\{I_\lambda ; \lambda \in \Lambda\}$ in $\mathcal{S}$

  upper bound $J = \bigcup_{\lambda \in \Lambda} I_\lambda \subsetneq R$

  Since $1 \notin I_\lambda \ \forall \lambda \Rightarrow 1 \notin J \Rightarrow J \neq R$

According to zorn lema the set $\mathcal{S}$ has the max element (maximal ideal) and it containes I

Cor:

R ring, $x \in R$ Then

$x \in R^\times \Leftrightarrow x$ does not lie in max ideal

Proof

$x \in R^\times \Leftrightarrow \langle x \rangle = R \quad \Rightarrow$

$x \in R \Leftrightarrow \langle x \rangle \subsetneq R \Leftrightarrow \exists M \overset{\max}{\subset} R \; x \in R \subseteq M \subsetneq R$

II.1 Factorisation

R ring, integral domain

u unit $\Leftrightarrow \exists u^{-1} \in R \Leftrightarrow \langle u \rangle = R$

$a|b \Leftrightarrow \exists g \in R.\ b = g \cdot a \Leftrightarrow b \in \langle a \rangle \Leftrightarrow \langle b \rangle \subseteq \langle a \rangle$

a and b are associates $a|b \wedge b|a \Leftrightarrow a = bu \Leftrightarrow \langle a \rangle = \langle b \rangle$

a is a proper devision of $b \Leftrightarrow b = ag$, $a, g \notin R^\times$

$\Leftrightarrow \langle b \rangle \subsetneq \langle a \rangle \subsetneq R$

a is irreducable $\Leftrightarrow a \notin R^\times$ and has no proper devisors

$\Leftrightarrow$ maximal among principa ideals

p is prime element $\Leftrightarrow p \in R^* \wedge (p|ab \Rightarrow p|a \vee p|b)$

$\Leftrightarrow \langle p \rangle$ is prime ideal

Definition Domain R is UFD (unique factorisation domain) if

1) each element is a product of irreducables and units
2) factorisation is unique up to units and order

Prop: An int-domain R is UFD $\Leftrightarrow$

1) every ascending chain of principal ideals stabilazes
2) every ireducable element is prime

Proof:

$\Rightarrow$. if $\langle a_1 \rangle \subset \langle a_2 \rangle \subset \dots$

then $a_{i+1} | a_i \wedge a_i / a_{i+1}$ is not a unit

So the numbers of factors strictly decreses

This canot happe so 1 holds

- Suppose p is ireducable and $p | ab$ Fedor a, b. The uniqueness of factorisation

Filipovi zapiski

Def: A domain R is principal ideal domain (PID) if every ideal is principal

Prop: R PID. Then

a) Every irreducible element is prime

b) Every nonzero ideal is max

Proof:

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III Radical ideals

Def: Jacobson radical of ring R $\operatorname{rad}(R)$ is the intersection of all max ideals of R

$$\operatorname{rad} R = \bigcap_{M \trianglelefteq R \text{ max}} M$$

Prop: R ring $I \trianglelefteq R$ $u \in R^\times$. Then

$$\forall x \in R. \left(x \in \operatorname{rad} R \iff \forall y \in R.\ u - xy \in R^\times \right)$$

In particular

- $R^\times + \operatorname{rad} R \subseteq R$
- $1 - I \subseteq R^\times \Rightarrow I \subseteq \operatorname{rad} R$

Proof: Filip

Cor: $I \trianglelefteq R$, $K: R \to R/I$ quotient homomorphism.
Assume $I \subseteq \operatorname{rad} R$. Then

$$K|_{\operatorname{Idem}(R)} : \operatorname{Idem}(R) \longrightarrow \operatorname{Idem} R/I \quad \text{is injective}$$

Proof:

Let $e, e' \in \operatorname{Idem}(R)$, $K(e) = K(e')$

$x := e - e'$ then $x^3 = e^3 - 3e^2e' + 3ee'^2 - e'^3 =$

$= e - 3ee' + 3ee' - e' = e - e' = x$

$x - x^2 = 0$

$x(1-x) = 0$ we the have $K(x) = 0 \Rightarrow x \in I$

$\Rightarrow x \in \operatorname{rad} R$

By the proposition $1 - x^2$ is invertible, because

$x \in \operatorname{rad} R \Rightarrow x = 0$ $(x(1-x^2) = 0 \Rightarrow x = 0)$

Def: Ring is **local** if it has only one maximal ideal

R is called **semi-local** if it has finitely maximal ideals. The **residue field** of a local field (R, M) is R/M

(M: max ideal)

Lema: R ring, $N = R - R^\times$ the set of non units. Then

R local $\Leftrightarrow$ $N \lhd R$

(because N contains all non units)

Proof:

($\Leftarrow$) $I \subsetneq R$ is contained in N, because I contains only non units. if $N \lhd R$ then this must be a maximal.

($\Rightarrow$) R is local with M maximal, then

$R - M = R^\times = R - N \Rightarrow M = N \Rightarrow N \lhd R$

∎

Example

$R := R' \times R''$ is not local, because $(1,1) = (1,0) + (0,1)$

Example: R ring, $n \in \mathbb{N}$. Formal power series (FPS)

In $X_1 \dots X_n$ are formal (oo) series of the form

$$\sum_{i \in \mathbb{N}^1} a_i X_1^{i_1} \cdot \dots \cdot X_n^{i_n} \qquad a_0 := \text{const. coef.}$$

We add and multiply the same as polinomials

$R[[X_1, \dots X_n]]$

$I := \langle X_1 \dots X_n \rangle \triangleleft R[[X]]$

Then $\sum a_i X_1^{i_1} \cdot \dots \cdot X_n^{i_n} \longmapsto a_0 \in R$ ($\in R[[X]]$)

is a surjective homomorphism with kernel I

$R[X]/I \cong R$

$M \triangleleft R$ an ideal, $N := I + MR[[X]] \triangleleft R[[X]]$

$\Rightarrow R[[X]]/N \cong R/M$

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$R[[X]]/N \cong R/M$

with isomorphism

$$\frac{R[[X]]}{I + MR[[X]]} \longrightarrow R/M$$

$$\sum a_i X^i \longmapsto$$

Prop: R ring. $S \subseteq R$ multiplicative set, $I \lhd R$
$I \cap S = \emptyset$.

Form $\mathcal{J} = \{ J \lhd R ; I \subseteq J \wedge J \cap S = \emptyset \}$

Then $\mathcal{J}$ contains maximal element and each such maximal element is a prime ideal

Proof: 1) $I \in \mathcal{J} \Rightarrow \mathcal{J} \neq \emptyset$

2) Let J_λ be a chain in $\mathcal{J}$. Define

$J = \bigcup J_\lambda \Rightarrow J$ is an ideal (because increasing chain of ideals is ideal)

Then $J \in \mathcal{J}$

$\Rightarrow J$ is an upper bound in $\mathcal{J}$

We use Zorn's lemma $\Rightarrow \mathcal{J}$ has maximal elements

Let $P \in \mathcal{J}$ be maximal element

P is prime.

Take $x, y \in R - P$. Then $P + \langle x \rangle$ and $P + \langle y \rangle$ aren't in $\mathcal{J}$ by maximality of P

This means $P + \langle x \rangle \cap S \neq \emptyset$

$\exists, p, q \in P. \exists a, b \in R.$ $p + ax \in S$ and $q + by \in S$

$\overset{S}{\underset{\text{multipl.}}{\Longrightarrow}}$ $(p+ax)(q+by) = \underbrace{pq}_{\in P} + \underbrace{pby}_{\in P} + \underbrace{qax}_{\in P} + \underbrace{axby}_{\in P} \in S$

$abxy \notin P$ because $P \cap S = \emptyset$

$\Rightarrow xy \notin P$

$\Rightarrow xy \in R - P$ ∎

Radical of ideals $I \subseteq R$

Its radical is $\sqrt{I} = \{x \in R ; \exists n \in \mathbb{N} ; x^n \in I\}$

- obviously $I \subseteq \sqrt{I}$, $\sqrt{\sqrt{I}} = \sqrt{I}$ and $J \subseteq I \Rightarrow \sqrt{J} \subseteq \sqrt{I}$

Def: $I \trianglelefteq R$ is radical $:\Leftrightarrow I = \sqrt{I}$

Lema:
if $I = \bigcap_\lambda P_\lambda$ P_λ prime, then I is radical.

Proof: Suppose $x \in \sqrt{I}$. Then $\exists n \in \mathbb{N}$
$x^n \in P_\lambda$ for all $\lambda \Rightarrow x \in P_\lambda \vee x \in P_\lambda \vee \ldots \Rightarrow x \in P_\lambda$
So $x \in \bigcap P_\lambda = I \Rightarrow x \in I$

Def: $nil(R) = \sqrt{\langle 0 \rangle}$; is the nil radical.
$I \trianglelefteq R$ is nilpotent $\Leftrightarrow \exists n \in \mathbb{N} . I^n = \langle 0 \rangle$

Since $radR = \bigcap_{M \trianglelefteq R} M$ is an intersection of prime ideals

- $radR$ is an radical ideal $\sqrt{radR} = radR$
- $\langle 0 \rangle \subseteq radR \Rightarrow \sqrt{\langle 0 \rangle} \subseteq \sqrt{radR}$

Def. ring R is reduced : if $nilR = \langle 0 \rangle$

Theorem: if $I \lhd R$, then $\sqrt{I} = \bigcap_{I \subseteq P \text{ prime}} P$

Proof. • let's say $x \notin \sqrt{I}$. Define $S = \{1, x, x^2, \ldots\}$

Then S is multiplicative and $S \cap I = \emptyset$

By the proposition the maximal among ideals of R that contain I and avoid S is a prime ideal

From $S \cap P = \emptyset$ we get $x \notin P \Rightarrow x \notin \bigcap_{I \subseteq P \text{ prime}} P$

• let say $x \in \sqrt{I}$ and $P \supseteq I$ is prime, then $x^n \in P$
$\overset{P \text{ prime}}{\Rightarrow} x \in P \Rightarrow x \in \bigcap R$

Corollary: If $I \trianglelefteq R$ then $\sqrt{I} \trianglelefteq R$

Proof: By theorem, $\sqrt{I}$ is the intersection of (prime) ideals, so it is an ideal ■

Alternative proof:

$R \cdot \sqrt{I} \subseteq \sqrt{I}$

$x, y \in \sqrt{I}$, say $x^n \in I$ $y^m \in I$

$(x+y)^{m+n-1} = \sum \binom{m+n-1}{k} x^k y^{m+n-1-k}$

if $k \geq n$ then $x^k \in I$

if $k < n$ then $m+n-1-k \geq m \Rightarrow y^{m+n-1-k} \in I$

$\Rightarrow x+y \in \sqrt{I}$

IV Modules

- R ring, R-module $\mathcal{M}$ is $(\mathcal{M},+)$ abelian with scalar multiplication $R\times\mathcal{M}\to\mathcal{M}$

$$(x,m)\mapsto x\cdot m$$

such that a) $x\cdot(m+n)=x\cdot m+xn$ } distributivity

b) $(x+y)\cdot m=xm+ym$

c) $(xy)m=x(ym)$ → asociativity

d) $1\cdot m=m$ ... monicity

A **submodul** is a sub group that is closed under scalar multiplication

$I\trianglelefteq R\Rightarrow I\cdot M$ is submodule of M generated by $x\cdot m\,;\,x\in I,\ m\in M$ modul!!!



- To each $m \in M$ we associate its annihilator
  $\mathrm{Ann}_R(m) := \{x \in R ; x \cdot m = 0\} \lhd R$
- $N \subseteq M \Rightarrow \mathrm{Ann}(M) \subseteq \mathrm{Ann}(N)$ and

Filp

• $\mu_x : \mathcal{M} \longrightarrow \mathcal{M} \qquad x \in R$

$\mu_x : m \longmapsto m \cdot x$

$\mu_R : R \longrightarrow End_{\mathbb{Z}}(\mathcal{M})$

$x \longrightarrow \mu_x$ is a ring homomorphism

$\ker \mu_R = Ann(\mathcal{M})$

conversly if $(N, +)$ is an abelian group

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- Algebras: A ring R' is R algebra if $\exists\ R \xrightarrow{\varphi} R'$
  M' R'-module becames R-module by restricting of
  scalars $\quad x \cdot m = \varphi(x) \cdot n$
  $R \quad M \quad R'$

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